

# Sebastian Mostek

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## EDUCATION

**CORNELL UNIVERSITY**  
**SIBLEY SCHOOL OF MECHANICAL AND AEROSPACE ENGINEERING**  
*Bachelor of Science in Mechanical Engineering*  
*Magna Cum Laude, Class of 2025*

Ithaca, NY  
August 2022 – December 2025  
GPA 3.9

## RELEVANT COURSEWORK

*Advanced Dynamics & Vibrations, Material Mechanics, Fluid Dynamics, System Dynamics, Heat Transfer, Quantum Mechanics, Differential Equations, Electric Circuits, Wave Physics, Mechanical Design, Probability & Statistics*

## PROFESSIONAL EXPERIENCE

**Private Machines**  
*Mechanical Engineer*

Brooklyn, NY  
April 2026 – September 2026

- Redesigned all sheet metal and machined components of the company's Enforcer Servers and server chassis to improve ease of assembly, DFM, and CAD editability.
- Led R&D into improving thermal contact, heat sink design, and chassis air-flow.
- Prepared thermal and CFD simulations in tandem with the planning of R&D experiments.
- Prototyped and 3D-printed various devices to improve assembly speed and reliability
- Trained assemblers and assisted in the work of building, testing, and shipping servers and chassis.
- Prepared designs for quotation by manufacturers and researched alternate suppliers

**PYE Education Center**  
*Math Tutor*

Elmhurst, NY  
June 2023 – September 2023

- Led one-on-one virtual tutoring sessions on high-school-level math
- Designed and reviewed SAT practice problems

## RESEARCH EXPERIENCE

**Laboratory for Advanced Materials and Manufacturing**  
*Researcher*

Ithaca, NY  
June 2025 – December 2025

- Developed a novel nondimensionalized theoretical model of the temperature distribution in the melt pool during Laser Power Bed Fusion with the aim of accelerating the identification of optimal process parameters for different materials
- Utilized advanced mathematical tools such as Finite Element Analysis and Fast Fourier Transforms to numerically evaluate the model
- Drafted a journal article about the model and its advantages over existing predictive models

**Independent Mathematics Research**  
*Researcher*

Ithaca, NY  
September 2022 – June 2025

- Enumerated and classified the equilibrium distributions of  $n$  mutually repulsive points confined to the surface of a sphere
- Studied advanced mathematical fields including Graph Theory, Group Theory, and Numerical Optimization
- Developed an elaborate algorithm in MATLAB spanning over 2500 lines of code to efficiently search for all possible solutions for a given  $n$

## SKILLS & INTERESTS

- **Professional skills:** CAD and FEA in Autodesk Fusion and Ansys; CFD; 3D printing; Programming in MATLAB and Python; GD&T; Microsoft Office Suite; LaTeX; basic machining
- **Interests:** Music; Hiking; Soccer; Board Games; Political History and Philosophy; Linguistics
- **Languages:** A2 German